

Projectile Motion

From Newton's second law to range, height and time of flight

Derivation notes — Claude Code session `meetingproject`, via RoboMeet — September 2026

1 Setup

A point mass m is launched from the origin with speed v_0 at angle θ above the horizontal. Gravity g acts downward; air resistance is neglected.

Governing equations (Newton's second law, per component)

$$\textbf{Horizontal: } m \ddot{x} = 0 \quad (1)$$

$$\textbf{Vertical: } m \ddot{y} = -mg \quad (2)$$

$$\textbf{Initial conditions: } x(0) = 0, y(0) = 0, \quad \dot{x}(0) = v_0 \cos \theta, \dot{y}(0) = v_0 \sin \theta \quad (3)$$

The two directions are *uncoupled*: gravity touches only y . Each equation integrates on its own.

2 Integrate twice

$$\dot{x}(t) = v_0 \cos \theta, \quad x(t) = v_0 \cos \theta t \quad (4)$$

$$\dot{y}(t) = v_0 \sin \theta - gt, \quad y(t) = v_0 \sin \theta t - \frac{1}{2}gt^2 \quad (5)$$

Trajectory: eliminate t

From (4), $t = x/(v_0 \cos \theta)$. Substitute into (5):

$$y(x) = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2 \quad \text{a parabola through the origin.} \quad (6)$$

3 Three results

Time of flight: $y = 0$ again

$$v_0 \sin \theta T - \frac{1}{2}gT^2 = 0 \implies T = \frac{2v_0 \sin \theta}{g} \quad (7)$$

Maximum height: $\dot{y} = 0$

The peak is reached at $t^* = v_0 \sin \theta / g = T/2$ (by symmetry). Then

$$H = v_0 \sin \theta t^* - \frac{1}{2}gt^{*2} \implies H = \frac{v_0^2 \sin^2 \theta}{2g} \quad (8)$$

Range: x at $t = T$

$$R = v_0 \cos \theta \cdot \frac{2v_0 \sin \theta}{g} \implies R = \frac{v_0^2 \sin 2\theta}{g} \quad (\sin 2\theta = 2 \sin \theta \cos \theta) \quad (9)$$

4 Observations

What the formulas say

- **Best angle.** R is largest when $\sin 2\theta = 1$, i.e. $\theta = 45^\circ$, giving $R_{\max} = v_0^2/g$.
- **Complementary angles.** θ and $90^\circ - \theta$ share the same range ($\sin 2\theta$ is symmetric about 45°) but not the same height or flight time.
- **Scaling.** Doubling v_0 quadruples both R and H ; doubling g halves them. Mass never appears.
- **Symmetry.** Rise and fall take equal time $T/2$; the speed at landing equals the launch speed.
- **Check.** $\theta = 90^\circ$: $R = 0$, $H = v_0^2/2g$, $T = 2v_0/g$ (straight up and down). $\theta = 0$: $T = 0$, nothing leaves the ground.